Assignment 2

Chen Baizheng 725/0800

Question 3: Weighted dominating set.

(a) use the path a-b-c, set $c_a = c_b = c_c = 1$, $c_b = L > 2$

Vertex b dominates all 3 vertices, each vertex dominates only 2

∴ The cost-blind rule picks b and pays $L$. The optimium picks a,c and pays 2.

⇒ The ratio $\frac{L}{2}$ can be arbitrarily large.

(b) When picks $v$, Let $Q = N[v] \cap U$

$\Rightarrow \sum_{u \in Q} p_u = |Q| \frac{c_v}{|Q|} = c_v$

$\Rightarrow \sum_{u \in S \cap V} p_u = c(D)$

(c) set $s = |N[v]| \Rightarrow |N[v] \cap U| \geq s - j + 1$

$\Rightarrow p_{u_j} = \frac{c_v}{|N[u_j] \cap U|} \leq \frac{c_v}{|N[v] \cap U|} \leq \frac{c_v}{s-j+1}$

$\Rightarrow \sum_{u \in N[v]} p_u \leq c_v \sum_{j=1}^{s} \frac{1}{s-j+1} = H_s c_v$

(d) $|N[v]| \leq \Delta + 1$

$c(D) = \sum_{u \in V} p_u \leq \sum_{v \in D^*} \sum_{u \in N[v]} p_u \leq \sum_{v \in D^*} H_{|N[v]|} c_v$

$\leq H_{\Delta+1} OPT$

Question 4 Steiner tree through a terminal MST

(a) $d(u,w) \le d(u,v) + d(v,w)$

(b) $c(A) \le c(\text{path union}) \le \sum_{uv \in M} d(u,v) = d(M)$

(d) $c(A) \le d(M) \le d(P)$
$\le (1-\frac{1}{k})L$
$\le 2(1-\frac{1}{k})c(T^*)$

For $k=2$, returns a shortest $u$-$v$ path

(c) $\sum_{i=1}^{k} d(r_i, r_{i+1}) \le 2d(T^*)$, $r_{k+1} = r_1$

(e) $t + (k-t)(2-\varepsilon) = k + (k-t)(1-\varepsilon) \ge k$

$\therefore OPT = k \Rightarrow 2-\varepsilon < 2$

costs $(k-1)(2-\varepsilon)$

$\Rightarrow \frac{c(A)}{OPT} = \frac{(k-1)(2-\varepsilon)}{k} \to 2(1-\frac{1}{k})$ $\varepsilon \to 0$

Question 5 Rent-or-buy with ~~cha~~ changing rental prices

(a) $OPT = \min\left\{B, \sum_{t=1}^{T} r_t\right\}$

(b) $ALG = 3+4+10 = 17$

$\Rightarrow OPT = \min\{10, 3+4+5+1\} = 10$

OPT buys on day 1 and ~~days~~ pays 1.

(c) $R = 0 < B \Rightarrow R + r_t < B$

all rents sum to $R < B$, so $ALG = R = OPT$

If it buys on day $T$, the $R + r_T \geq B$

$OPT = B \Rightarrow ALG = R + B < 2B = 2OPT$

2-comptitve

(d) $r_t = B/M$, rents days of $M-1$

$$ALG = \frac{M-1}{M}B + B,$$

$$OPT = B$$

$$\frac{ALG}{OPT} = 2 - \frac{1}{M}$$

(e) $T \leq M$ $OPT = TB/M$ $ALG = B + (T-1)B/M$

$$\therefore \frac{ALG}{OPT} = 1 + \frac{M-1}{T} \geq 2 - \frac{1}{M}$$

$T > M$, $OPT = B$ $\frac{ALG}{OPT} = 1 + (T-1)M \geq 2$

Never buy $T = 2M$ $\frac{ALG}{OPT} = 2$

As $M \to \infty$ get 2